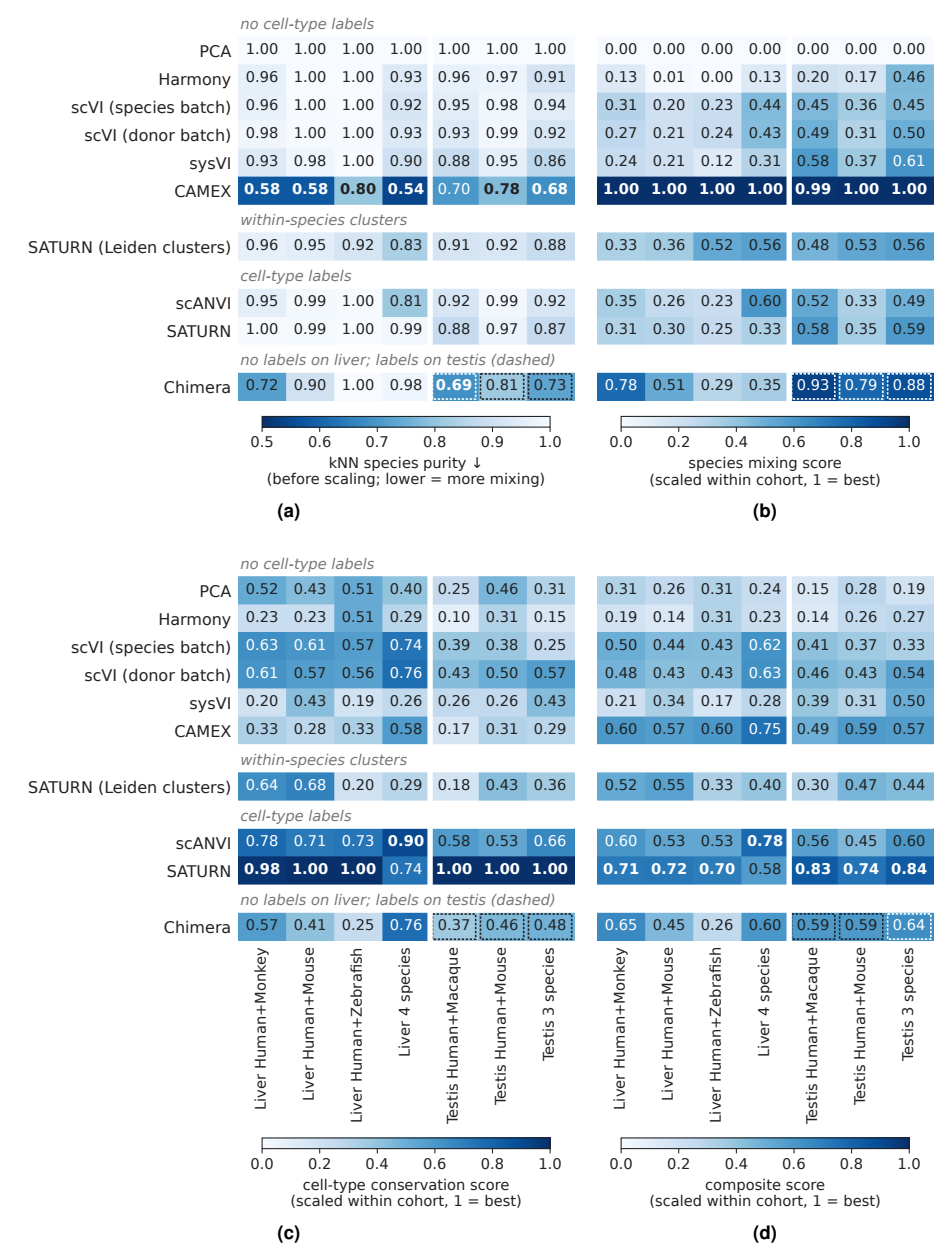

no cell-type labels
PCA 1.00 1.00 1.00 1.00 1.00 1.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Harmony 0.96 1.00 1.00 0.93 0.96 0.97 0.91 0.13 0.01 0.00 0.13 0.20 0.17 0.46
scVI (species batch) 0.96 1.00 1.00 0.92 0.95 0.98 0.94 0.31 0.20 0.23 0.44 0.45 0.36 0.45
scVI (donor batch) 0.98 1.00 1.00 0.93 0.93 0.99 0.92 0.27 0.21 0.24 0.43 0.49 0.31 0.50
sysVI 0.93 0.98 1.00 0.90 0.88 0.95 0.86 0.24 0.21 0.12 0.31 0.58 0.37 0.61
CAMEX 0.58 0.58 0.80 0.54 0.70 0.78 0.68 1.00 1.00 1.00 1.00 0.99 1.00 1.00
within-species clusters
SATURN (Leiden clusters) 0.96 0.95 0.92 0.83 0.91 0.92 0.88 0.33 0.36 0.52 0.56 0.48 0.53 0.56
cell-type labels
scANVI 0.95 0.99 1.00 0.81 0.92 0.99 0.92 0.35 0.26 0.23 0.60 0.52 0.33 0.49
SATURN 1.00 0.99 1.00 0.99 0.88 0.97 0.87 0.31 0.30 0.25 0.33 0.58 0.35 0.59
no labels on liver; labels on testis (dashed)
Chimera 0.72 0.90 1.00 0.98 0.69 0.81 0.73 0.78 0.51 0.29 0.35 0.93 0.79 0.88
0.5 0.6 0.7 0.8 0.9 1.0
kNN species purity ↓
(before scaling; lower = more mixing)
(a)
0.0 0.2 0.4 0.6 0.8 1.0
species mixing score
(scaled within cohort, 1 = best)
(b)
no cell-type labels
PCA 0.52 0.43 0.51 0.40 0.25 0.46 0.31 0.31 0.26 0.31 0.24 0.15 0.28 0.19
Harmony 0.23 0.23 0.51 0.29 0.10 0.31 0.15 0.19 0.14 0.31 0.23 0.14 0.26 0.27
scVI (species batch) 0.63 0.61 0.57 0.74 0.39 0.38 0.25 0.50 0.44 0.43 0.62 0.41 0.37 0.33
scVI (donor batch) 0.61 0.57 0.56 0.76 0.43 0.50 0.57 0.48 0.43 0.43 0.63 0.46 0.43 0.54
sysVI 0.20 0.43 0.19 0.26 0.26 0.26 0.43 0.21 0.34 0.17 0.28 0.39 0.31 0.50
CAMEX 0.33 0.28 0.33 0.58 0.17 0.31 0.29 0.60 0.57 0.60 0.75 0.49 0.59 0.57
within-species clusters
SATURN (Leiden clusters) 0.64 0.68 0.20 0.29 0.18 0.43 0.36 0.52 0.55 0.33 0.40 0.30 0.47 0.44
cell-type labels
scANVI 0.78 0.71 0.73 0.90 0.58 0.53 0.66 0.60 0.53 0.53 0.78 0.56 0.45 0.60
SATURN 0.98 1.00 1.00 0.74 1.00 1.00 1.00 0.71 0.72 0.70 0.58 0.83 0.74 0.84
no labels on liver; labels on testis (dashed)
Chimera 0.57 0.41 0.25 0.76 0.37 0.46 0.48 0.65 0.45 0.26 0.60 0.59 0.59 0.64
Liver Human+Monkey
Liver Human+Mouse
Liver Human+Zebrafish
Liver 4 species
Testis Human+Macaque
Testis Human+Mouse
Testis 3 species
0.0 0.2 0.4 0.6 0.8 1.0
cell-type conservation score
(scaled within cohort, 1 = best)
(c)
0.0 0.2 0.4 0.6 0.8 1.0
composite score
(scaled within cohort, 1 = best)
(d)